

04/24/2024

Purchase Order: 26627161

Heat Number

24001782

Item Number

11000102

Certificate of Tests

Sales Order: 1001000647

Vendor Name

Big River Steel - Osceola AR US

Mfg. Date

03/14/2024

PAGE: 11

Description

5L-X52 4-1/2X.337 (15.00#)BR,DRL,PEB

CHEMICAL ANALYSIS

	C	Mn	P	S	Ni	Cr	Cu	Mo	Si	Ti	V	CE
Ladle	0.0500	0.5200	0.0080	0.0020	0.0600	0.0900	0.1500	0.0130	0.1500	0.0010	0.0020	0.0956
	Al:	0.0330	Sn:	0.0060	N:	0.0064	B:	0.0001	Ca:	0.0023	Nb:	0.0120
Chem	0.0837	0.5499	0.0071	0.0021	0.0692	0.0814	0.1264	0.0121	0.1479	0.0008	0.0038	0.1291
	Al:	0.0347	Sn:	0.0052	N:	0.0150	B:	0.0000	Ca:	0.0023	Nb:	0.0133
Chem	0.0627	0.5465	0.0074	0.0018	0.0684	0.0815	0.1262	0.0116	0.1547	0.0009	0.0042	0.1080
	Al:	0.0385	Sn:	0.0050	N:	0.0151	B:	0.0000	Ca:	0.0025	Nb:	0.0140

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 67.60 TS (ksi): 72.50

EL (%): 31.00

HRB: 90.00

Y/T(%): 0.93

CHARPY TEST RESULTS

BODY

ORIENTATION: TRANSVERSE

Spec. Size (in) : .394"X.197"X2.165"

Test Temp,(of) -20

Location 3T90

Specimen(S)

Energy (ft/lbs.)

%Shear

1

72.00

100.00

2

75.00

100.00

3

78.00

100.00

Average

75.00

100.00

Acceptance Criteria Min. FT/lbs : Min. Ft/Lbs 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Big River Steel - Osceola AR US
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept